ZEAL COLLEGE OF ENGINEERING & RESEARCH, PUNE
PRESENTS

RoboMaze

Offical Event Rulebook

AVISHKAR '26
WHERE CURIOSITY BECOMES CREATION

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1. Introduction:

The Maze Solver Robot Competition challenges teams to build an autonomous robot that can navigate a maze and reach the exit without human help. It tests robot design, algorithm efficiency, sensor integration, and decision-making.

Participants develop intelligent robots that analyze surroundings and choose the fastest path in real time using robotics and control system concepts.

With multiple paths and dead ends, the maze demands adaptability, making the competition a test of both speed and smart problem-solving.

2. Eligibility of participation

- The competition accepts participation of teams and not individuals.
- The team consists of 2-4 persons.
- All U.G Students allowed for this competition and Only one robot per team is allowed.

3. The Robot

- The robot shall be self-contained (no remote controls).
- Robot must contain its own power supply and processor.
- The robot shall not leave any part of its body behind while negotiating the maze.
- The robot shall not jump over, fly over, climb, scratch, cut, burn, mark, damage, or destroy the walls of the maze.
- The robot shall not be larger, either in length or in width, than 30 centimeters. dimensions of a robot that changes its geometry during a run shall not be greater than 30 cm × 30 cm There are no restrictions on the height of the robot.
- **Weight Criteria**: Maximum 1 Kg

4. The Maze

- The maze of the competition remains secret until the competition day. Participants are expected to take into consideration the details below and develop a generic code that can perform successfully on any maze.
- The maze is composed of 40 cm × 40 cm unit squares and comprises up to 1x9 unit squares. The walls of the maze are 15 cm high and 6 mm thick (assume 5 % tolerance for mazes). Thus, the distance from wall to wall within a square is 40 cm. The outside wall encloses the entire maze.
- The sides of the maze walls are white, the tops of the walls are red, and the floor is black finished with matt colour.
    - Warning: Do not assume the walls are consistently white, or that the tops of the walls are consistently red, or that the floor is consistently black. Fading may occur; parts from different mazes may be used. Do not assume the floor provides a given amount of friction.
    - The start of the maze is located at one of the four corners. The start square is bounded on three sides by walls. The start line is located between the first and second squares. The destination goal is the five 40 cm × 40 cm cells at the center of the maze. The finish line is at the entrance to the destination square.
    - Multiple paths to the destination square are allowed and are to be expected. The destination square has only one entrance and it will be positioned so that a wall-hugging mouse will not be able to find it.
    - Maze layout will be revealed 15 minutes before the competition..

5. Entry Fees:

- ₹ 499/- per team

6. Arena

Round 1:

Total Cells	9 X 9 cells (each cell of 39.7cm) = 81 cells
Wall Thickness	6 mm

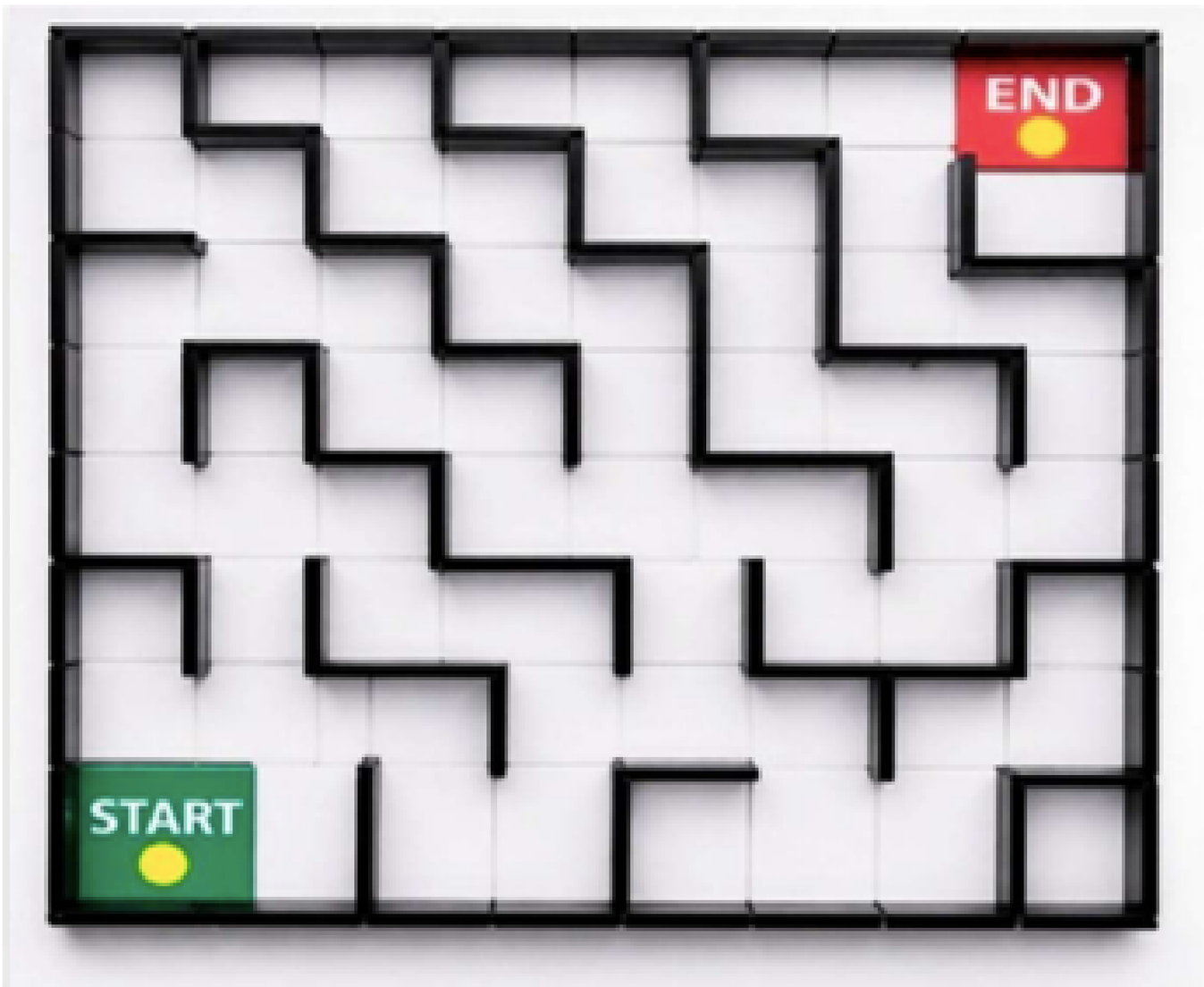
Round 2:

Total Cells	11 X 11 Cells (each cell of 39.7cm) = 121 cells
Wall Thickness	6 mm

Note :- the arena has one exit and two paths one is shortest path and another is longest. so code your robot as intelligent bot using shortest distance planning algorithm

7. The Competition

- 1. Each contesting robot is allocated a total of 12 minutes of access to the final maze. Any time used to adjust a robot between runs is included in the 12 minutes. Each run (from the start cell to the End zone) in which a robot successfully reaches the destination square is given a run time. The minimum run time shall be the robot's oƒicial time. First prize goes to the robot with the shortest official time. Second prize to the next shortest, and so on. ranked based on maximum no of checkpoints achieved
- 2. Each run shall be made from the starting square. The operator may abort a run at any time. If an bot touches the wall during a run the team will be given 3 attempts to keep robot at correct place the before checkpoint, it is deemed aborted, or the robot must be removed from the maze. If a robot has already crossed the finish line, it may be removed at any time without affecting the run time of that run.
- 3. After the maze is disclosed, the operator shall not feed information on the maze into the robot.
- 4. The contestant is allowed to:
  - change switch positions;
  - adjust sensors;
  - make repairs in case the robot breaks down.
- 5. The run timer will start when front edge of the robot crosses the start line and stops when the front edge of the robot crosses the finish line.
- 6. Sample Areana for Reference :



ROUND 1:

- Robot must navigate from START → END
- Focus on basic navigation + stability
- Simple maze (less complex)
- Includes Checkpoints: P1, P2, P3, P4
- Maximum time: 12 minutes per team
  (2 Minutes for readiness)
- Multiple runs allowed within time
- Best run time will be considered
- Reaching END get bonus Points
- Maze Consist of total checkpoint (P1, P2, P3,P4,)
- Maximum 3 restarts allowed
- Restart from previous valid checkpoint

Teams ranked based on: a. Highest score
b. Lowest time

ROUND 2:

- More difficult maze
- Hidden paths / dead ends
- Checkpoints may or may not be present
- Multiple runs allowed (maximum-3)
- Best run counted
- Reaching END gets bonus Points
- Total checkpoint P1, P2, P3, P4, P5
- Maximum time: 15 minutes per team (2 Minit for readiness)
- Ranking mainly based on : i. Fastest time Accuracy
  ii. Highest score
  iii. Fastest completion time
  vi. Least penalties

8. Terms and Conditions of Participation

- Participation in ROBOMAZE SOLVER assumes and requires acceptance of all terms and conditions for participation by competitors, the coaches and the organisations they represent.
- Technical inspection of the robot, marking the robot with a number sticker, and the order in which it will compete which is generated by an algorithm in the information system.
- All questions and issues that may arise during the competitions must be reported to the judges.
- The final decision about objections will be taken by the judges in cooperation with the organisers.
- Judge's decisions on any objections are considered final and can't be challenged by participants, the coaches or the organisations they represent

9. Robot Technical Control

- The robot's technical control will take place on the day of the competition at an area and on time specified by the organisers.
- Failure of a team to come in time for a robot's technical check leads to the team being excluded from the event.
- Technical control takes place before each phase of the competition (preliminary, qualifying, final) in which the team may participate
- Technical control includes the control of the robot based on the above and the paragraph "The Robot", if this paragraph exists. If the robot does not meet the requirements, it will not be accepted to compete and will automatically be disqualified from the event.
- The leader of the team only is responsible to take the team's robot for technical control.

10. Scoring Overview

1	If robot Choose Shortest path then bonus	**5**
2	If robot solves the maze without touching any single time to walls then	5
3	Shortest time taken by robot	5
4	Damage to Walls of maze by robot (negative marks)	10
5	Each Checkpoint covered will be awarded	10
6	Maze completed successfully	10

Note:- Participants who complete the maze using the shortest path shall receive full marks for all checkpoints with bonus points.
~ All the points converted into percentage for final Result ~

11. Judging Criteria

No.	Criteria	Weightage
1	Accuracy & Precision	25%
2	Adaptability	25%
3	Speed and efficiency	10%
4	Design and Construction	15%
5	Technical Implemention	15%

Tie-Breaker

1.More checkpoints
2.Faster final run
3.Fewer penalties

12. Changes and Cancellation of Rules

Any changes and/or cancellations in the rules of the competition are decided by the Robo-Maze solving in consultation with the Organizing Committee of the AVISHKAR

13. Prize Pool

1st Place: ₹ 20,000

2nd place: ₹ 15,000

3rd place: ₹ 10,000

14. Co-ordinators Contact

Student Co-ordinator	Aniket Nagare	7666604125
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